SUPPORTING ADULT GUIDANCE

The expression behind it

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The learner expands a product of two linear brackets, $(x + a)(x + b)$, using an **area model**: a rectangle split into four regions — x^2, the two x-strips (ax and bx), and the constant ab — whose areas sum to $x^2 + (a + b)x + ab$. The core outcome for everyone is **expanding** (add for the middle term, multiply for the constant), understood through the picture rather than a memorised rule. A **stretch tier** — **factorising** a simple quadratic $x^2 + bx + c$ — runs the same model backwards: given the area, find the two side-numbers that multiply to c and add to b. Negatives, and the case $a \neq 1$, are deliberately out of scope for this first meeting.

Driving Question: *What does x^2 look like as an area — and can we build it and take it apart?*

The mathematics, in plain terms

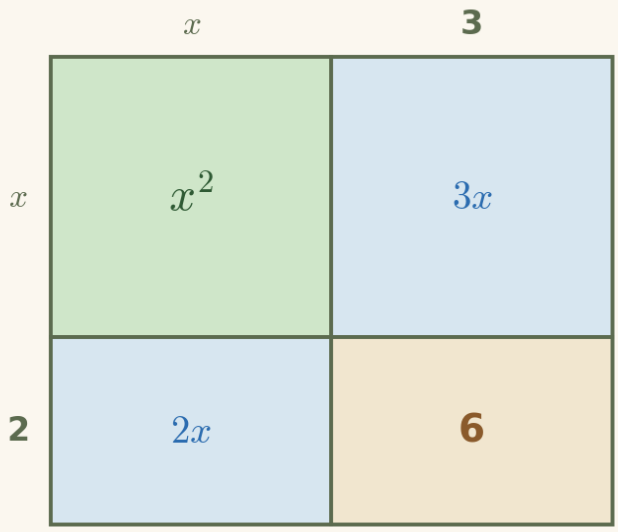
This lesson gives a pair of brackets a picture. Because the area of a rectangle is width × height, **$(x + 2)(x + 3)$** is the area of a rectangle whose sides are $(x + 2)$ and $(x + 3)$. Split each side where the x ends and the number begins and the rectangle falls into four pieces: a big x **by** x **square** (area x^2), a strip that is x **by 3** (area $3x$), a strip that is **2 by** x (area $2x$), and a small **2 by 3** rectangle (area 6). Add them: $x^2 + 3x + 2x + 6 = x^2 + 5x + 6$. Two things fall straight out of the picture and are worth naming aloud. The **middle term is the two numbers added** (the two strips, $3x$ and $2x$, join to $5x$), and the **constant is the two numbers multiplied** (the little corner, $2 \times 3 = 6$). Almost every early error is one of these two confused — multiplying to get the middle, or adding to get the constant — and the rectangle catches both. The great advantage of the area model over a mnemonic (like FOIL) is that it means something: a learner who forgets a rule can always redraw the rectangle. The **stretch** is to read the picture backwards — **factorising**. Given $x^2 + 7x + 12$, you are given the area and asked for the sides: which two numbers **multiply to 12 and add to 7**? (3 and 4, so it is $(x + 3)(x + 4)$.) Keep the numbers small and positive; the aim is the idea, not heavy manipulation.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — expand with the area model

$(x + 2)(x + 3)$ is the area of this rectangle. Read the four pieces, then join the two x-strips into one middle term.

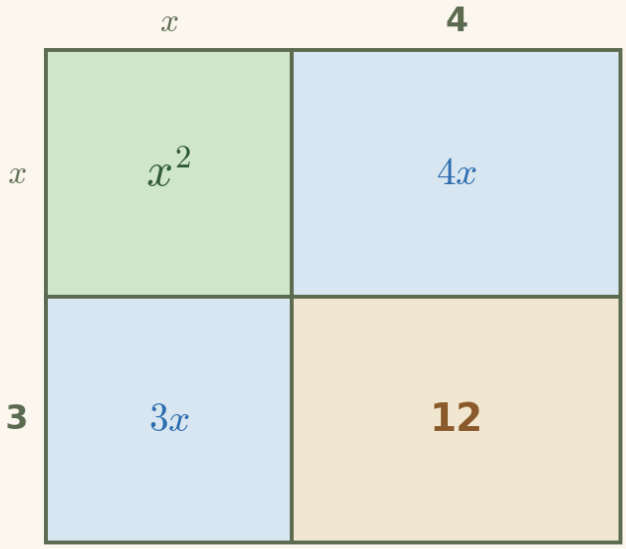


1. the square x^2
2. the two strips join (add) $3x + 2x = 5x$
3. the corner (multiply) $2 \times 3 = 6$

$(x + 2)(x + 3) = x^2 + 5x + 6$

Example 2 — factorise — the model backwards (stretch)

Now you are given the area, $x^2 + 7x + 12$, and you find the sides. You need two numbers that multiply to 12 and add to 7.



1. pairs that multiply to 12 — 1×12, 2×6, 3×4
2. which pair adds to 7? — 3 + 4 = 7
3. so the sides are — $(x+3)(x+4)$

$x^2 + 7x + 12 = (x + 3)(x + 4)$

Draw the rectangle

The organising move of this lesson is simply to **draw the rectangle** — sides (x + a) and (x + b), split into four labelled areas — and read it. Insist on the picture while it is still doing work; the four areas make the structure of a quadratic visible in a way a rule never will. From the rectangle, two facts do everything: the **middle term is a + b** (the two strips joining) and the **constant is a × b** (the corner). Name these each time, because the two classic slips are exactly these two confused. When a learner is secure with expanding, offer the **reverse** as a stretch: hand them the area (the quadratic) and ask for the sides. Factorising is not a new technique to memorise but the same rectangle read the other way — which is precisely why building it forwards first matters so much. Keep everything small and positive here; the general and negative cases belong to later work. Praise the learner who redraws the rectangle when unsure — that is the transferable habit, not a remembered rule.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing

Connection	The rectangle behind the brackets: build the area model for (x + a)(x + b) and read its four pieces.	~10 min
Movement	Expand it: expand products to x^2 + (a + b)x + ab, with the area model on hand (dominant).	~14 min
Reflection	Spot the slip: catch the two classic errors — multiplying for the middle, or mishandling the constant.	~10 min
Creativity	Factor Finder (stretch): run the model backwards to factorise a simple quadratic.	~7 min
Rest	A pause after turning a rule into a picture with a reason.	~2 min
Nutrition	Mode B (intellectual): what the learning feeds — the power of a good representation to make an abstract symbol make sense.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Multiplying the two numbers to get the middle term

Repair: “The two strips join to make the middle term, so you add: 2 + 3 = 5. Multiplying gives the corner, not the middle.”

Adding the two numbers to get the constant

Repair: “The corner is the two numbers multiplied: 2 × 3 = 6, not 2 + 3. Add for the middle, multiply for the end.”

Forgetting the middle term altogether

Repair: “There are four pieces, not two. Where are the two strips? Their areas make the middle term.”

Writing $x + x$ for the x^2 square

Repair: “The square is x times x, which is x^2, not $x + x$. Its sides are both x.”

Factorising: finding numbers that add but do not multiply (or vice versa)

Repair: “Both must be true at once — the pair has to multiply to the constant AND add to the middle. List the pairs that multiply, then find the one that adds correctly.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the area of a rectangle, multiplying out a single bracket (the distributive law), and collecting like terms. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner reading the rectangle:

- “Draw the rectangle. What are the four areas?”
- “The two strips — what do they add up to? That is your middle term.”
- “The little corner — what is 2 × 3? That is your constant.”
- “Stretch: which two numbers multiply to 12 and add to 7?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“Why use an area model instead of just a rule like FOIL?”

Because the rectangle *means* something. A learner who forgets a rule is stuck, but a learner who can draw the rectangle can always rebuild the answer — and the picture makes the middle-term and constant come out correctly by construction.

“Is factorising expected of everyone?”

No — expanding is the core outcome for all. Factorising is offered as a stretch for learners who are secure with expanding, and it is simply the same area model read backwards.

“What about negative numbers, or x^2 with a number in front?”

Both are out of scope for this first meeting, to keep the picture clean. They are handled later in the pathway; here the numbers stay small and positive.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.

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